Earthquake_Project – Presentation Script

Slide 1: Introduction

Good morning/afternoon everyone.

My project is **"Global Seismic Trends: Data-Driven Earthquake Insights."** The objective of this project is to collect, process, analyze, and visualize global earthquake data to identify seismic patterns and provide meaningful insights for disaster management and research.

Slide 2: Problem Statement

Earthquakes occur frequently around the world and generate a large amount of seismic data. Analyzing this data manually is difficult and time-consuming. This project helps organize earthquake information, identify trends, and present the results through an interactive dashboard for easier analysis.

Slide 3: Objectives

The main objectives of this project are:

- Collect earthquake data from the USGS Earthquake API.
- Clean and preprocess the raw data.
- Store the processed data in a MySQL database.
- Perform SQL-based analysis.
- Develop an interactive Streamlit dashboard.
- Generate insights from earthquake trends and patterns.

Slide 4: Technology Stack

The project was developed using:

- Python for data collection and preprocessing.
- Pandas for data manipulation.
- MySQL and SQLAlchemy for database storage.
- SQL for analytical queries.
- Streamlit for dashboard development.
- Matplotlib and Seaborn for data visualization.

Slide 5: Project Workflow

The workflow consists of the following steps:

1. Data Collection using the USGS API.
2. Data Cleaning and Preprocessing.
3. Feature Engineering.
4. MySQL Database Storage.
5. SQL Query Analysis.
6. Dashboard Development.
7. Business Insights and Reporting.

Slide 6: Data Preprocessing

During preprocessing:

- Missing values were handled.
- Duplicate records were removed.
- Data types were standardized.
- Date and time fields were converted into readable formats.
- New columns such as Year, Month, Day, Country, and Depth Category were created for better analysis.

Slide 7: SQL Analysis

Several SQL queries were performed to analyze:

- Top 10 strongest earthquakes.
- Deepest earthquakes.
- Country-wise earthquake counts.
- Monthly earthquake trends.
- Average earthquake magnitude.
- Tsunami-related events.
- Magnitude and depth distributions.

Slide 8: Dashboard Demonstration

The Streamlit dashboard provides:

- Total earthquake count.
- Magnitude distribution.
- Country-wise earthquake analysis.
- Monthly trend visualization.
- Depth category analysis.
- Tsunami event statistics.
- Interactive filters for data exploration.

Slide 9: Key Insights

The major findings are:

- Most earthquakes occur at shallow depths.
- Moderate-magnitude earthquakes are more common than high-magnitude events.
- Seismic activity varies significantly across countries and over time.
- Interactive dashboards make earthquake trend analysis easier and support informed decision-making.

Slide 10: Conclusion

This project successfully demonstrates the complete data analytics process, from collecting real-world earthquake data to preprocessing, database storage, SQL analysis, and dashboard visualization. It provides valuable insights into global seismic activity and can be extended with real-time monitoring and predictive analytics in the future.

Thank you for your time. I am happy to answer any questions.